

The Language of Medicine

13th edition

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Chapter 15

Musculoskeletal System

Chapter Goals (Slide 1 of 3)

- Define terms relating to the structure and function of bones, joints, and muscles.
- Describe the process of bone formation and growth.
- Locate and name the major bones of the body.

Chapter Goals (Slide 2 of 3)

- Analyze the combining forms, prefixes, and suffixes used to describe bones, joints, and muscles.
- Explain various musculoskeletal disease conditions and terms related to bone fractures.

Chapter Goals (Slide 3 of 3)

- Describe important laboratory tests and clinical procedures related to the musculoskeletal system, and recognize relevant abbreviations.
- Apply your new knowledge to understanding medical terms in their proper contexts, such as medical reports and records.

Chapter 15

Lesson 15.1

Introduction

Musculoskeletal System: bones, muscles, and joints of the body

Bones: structural support and protection of internal organs

Muscles: internal and external movement

Joints: where bones come together/type determined by need for flexibility

Tendons: bind muscles to bones

Ligaments: bind bones to other bones

Physicians Associated with Musculoskeletal System

- **Orthopedist: MD**
- **Rheumatologist: MD**
- **Physiatrist: MD**
- **Osteopathic physician: DO**

Bone Formation (Slide 1 of 3)

Ossification: replacement of cartilage with bone

- **Cells:**

- **Osteoblast:** produce immature bony tissue that replaces cartilage
- **Osteocyte:** nourishes and maintains bone
- **Osteoclast:** reabsorb or digest bone (remodels bone)

Bone Formation (Slide 2 of 3)

- Osteoblasts and osteoclasts work to deposit and tear down bone throughout life.
- The skeleton is a source of calcium.
- Proper formation of bone depends on sources of calcium, phosphorous, and vitamin D.
- Calcium phosphate enzyme helps create hard bone from these sources.

Bone Formation (Slide 3 of 3)

- Reservoir for calcium storage
- Calcium necessary for nerve transmittal to muscle, including heart muscle and muscles attached to bones
- Calcium level maintained in blood by parathyroid gland, which secretes a hormone to release calcium from bone

Bone Structure (Slide 1 of 4)

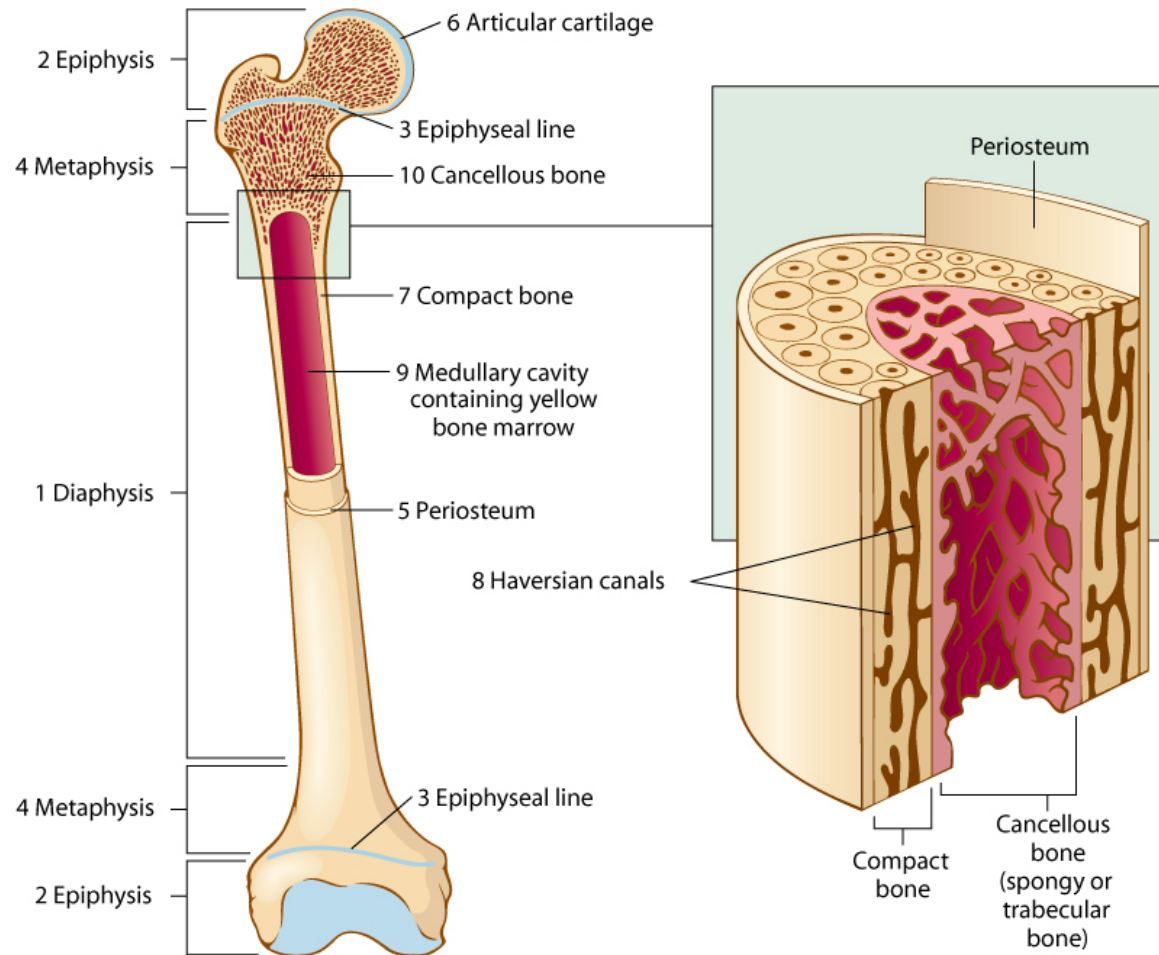
- Long, short, flat, sesamoid (in shape)
- **Diaphysis:** shaft
- **Epiphysis:** end
- **Metaphysis:** conelike flared portion between end and shaft
- **Epiphyseal line or plate:** growth plate where cartilage replaced by bone for bone growth (in length)

Bone Structure (Slide 2 of 4)

Interior bone structure includes:

- Articular cartilage
- Epiphyseal line or plate
- Cancellous bone
- Compact (cortical) bone
- Medullary cavity
- Periosteum

Bone Structure (Slide 3 of 4)



Bone Structure (Slide 4 of 4)

- Ends of bones covered by articular cartilage in the joint: cushions joint and allows it to move smoothly
- **Compact bone:** contains haversian systems for blood vessels, nerves, and yellow bone marrow (mostly fat)
- **Cancellous** (spongy/trabecular bone): spaces contain red bone marrow with elements for blood formation

Bone Marrow

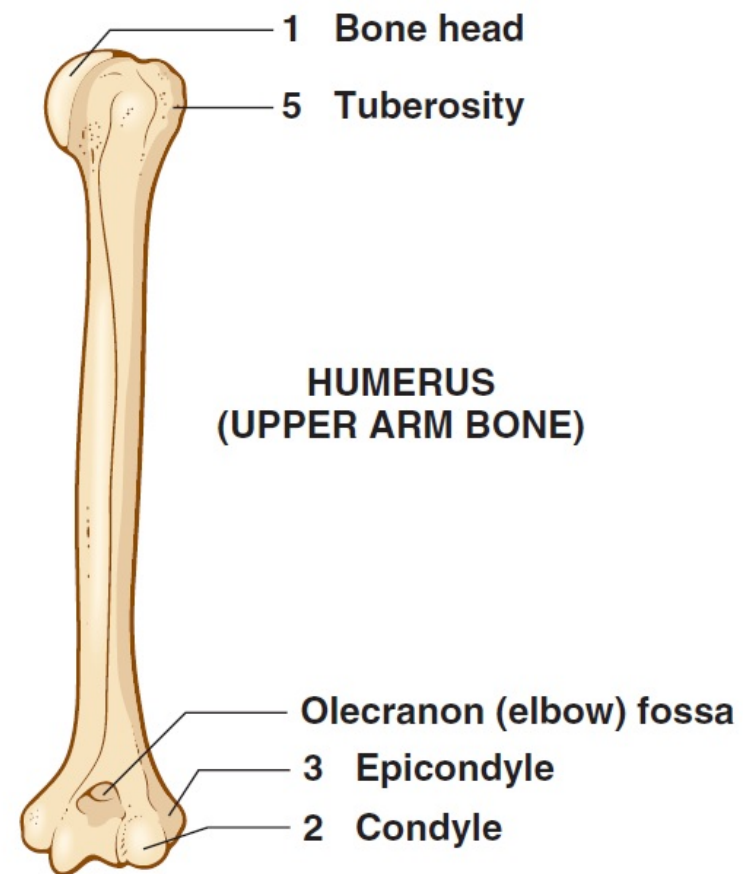
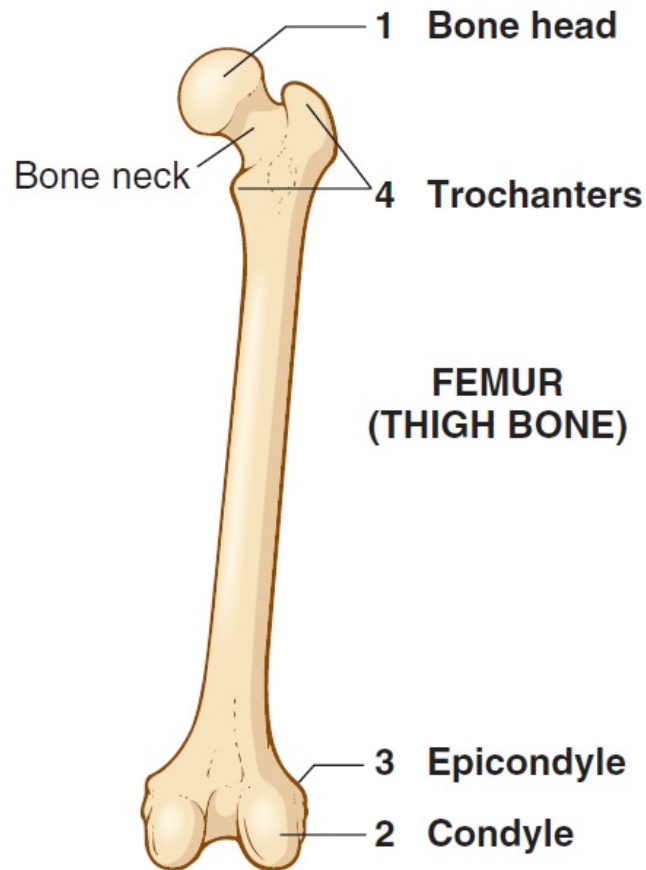
- **Yellow marrow:** chiefly fat
- **Red marrow:** rich with blood vessels and immature and mature blood cells in various stages of development; in later life replaced with yellow marrow; **hematopoieses** is the production of all types of blood cells in the bone marrow

Bone Processes and Depressions

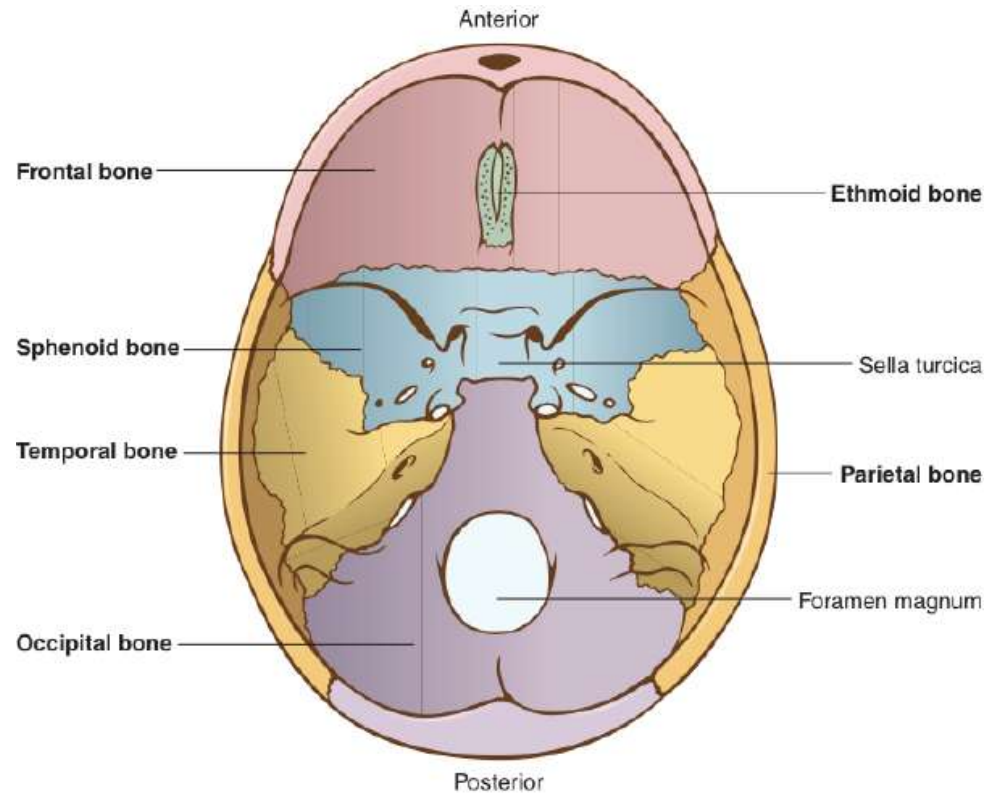
(Slide 1 of 2)

- **Processes:** serve as attachments for muscles, tendons, and ligaments
- **Depressions:** openings or hollow regions help join bones or serve as passageways for nerves and/or vessels

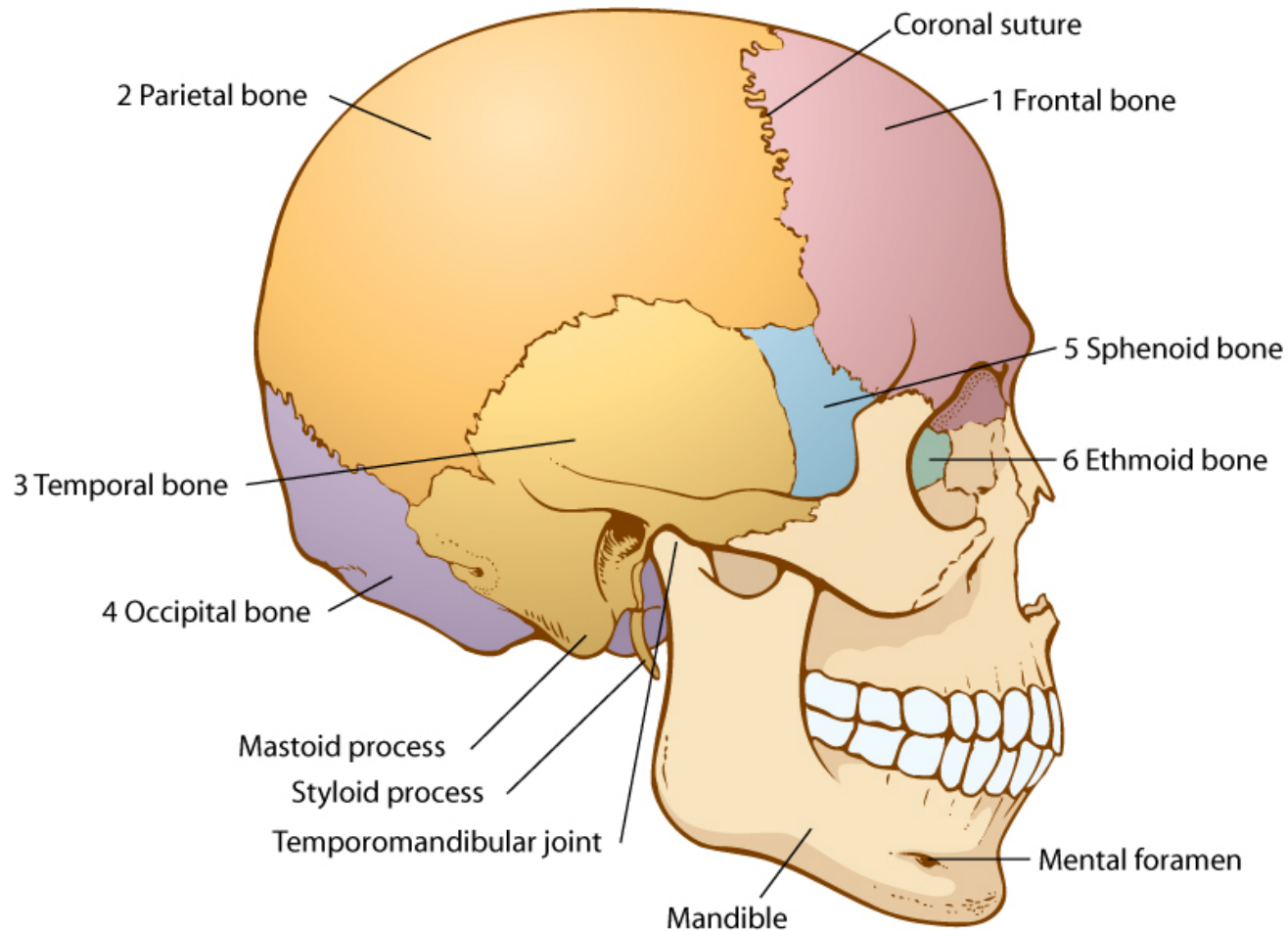
Bone Processes and Depressions (Slide 2 of 2)



Cranial Bones (Slide 1 of 2)



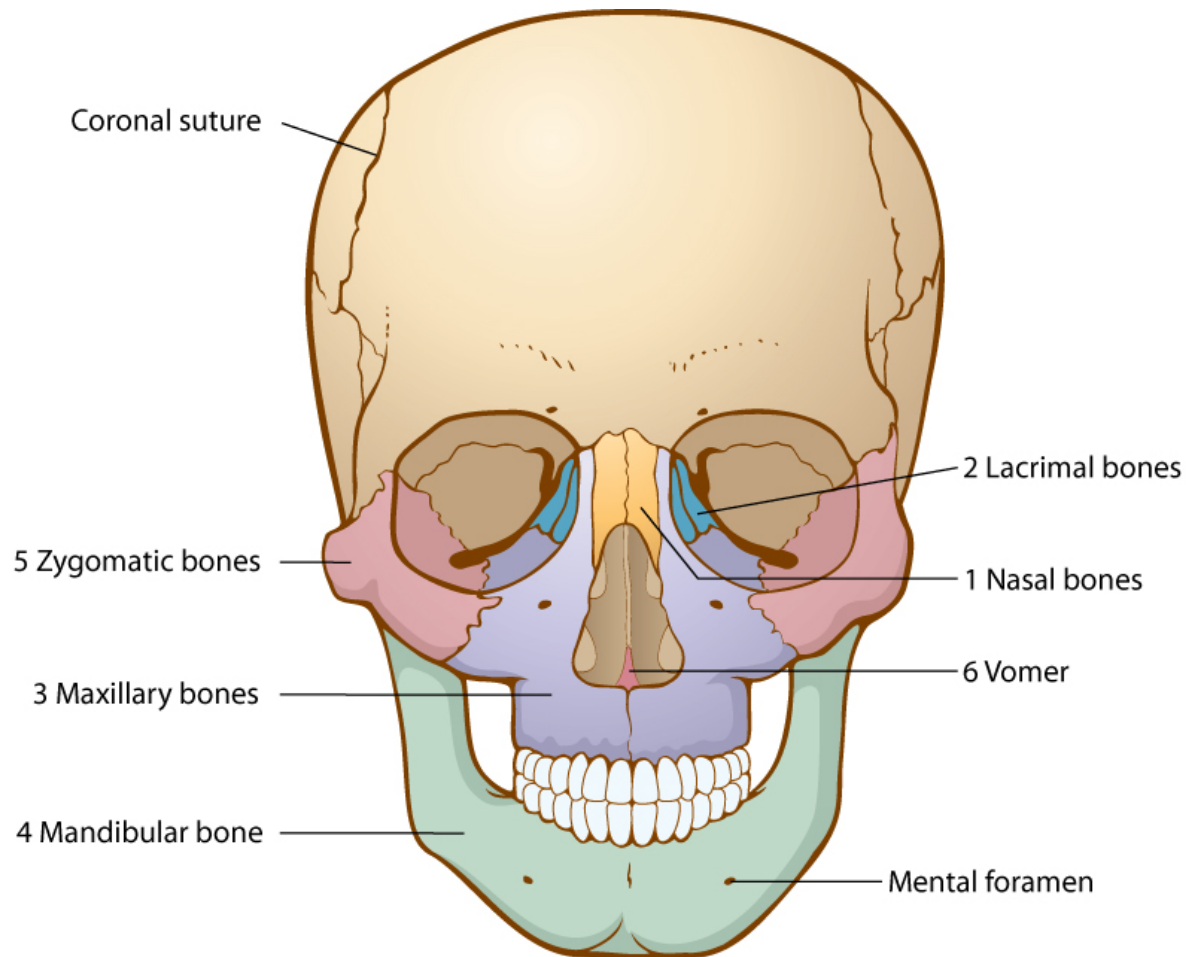
Cranial Bones (Slide 2 of 2)



Cranial Bones and Associated Structures

- **Frontal bone**
- **Parietal bone**
- **Temporal bone**
 - temporomandibular (TMJ) joint
 - mastoid process
 - styloid process
- **Occipital bone**
 - foramen magnum
- **Sphenoid bone**
 - sella turcica
- **Ethmoid bone**

Facial Bones (Slide 1 of 2)



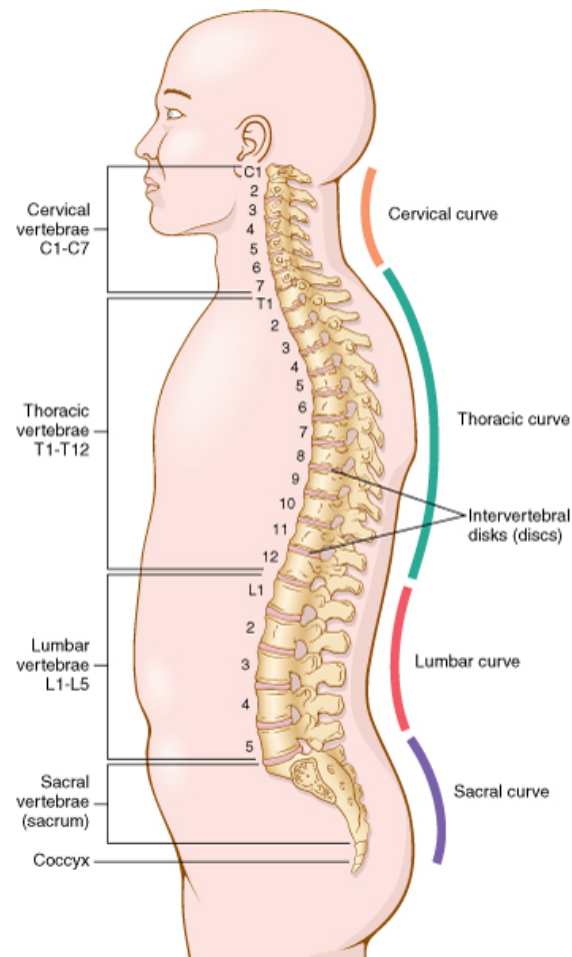
Facial Bones (Slide 2 of 2)

- Nasal bones
- Lacrimal bones
- Maxillary bones
- Mandibular bones
- Zygomatic bones
- Vomer

Sinuses

Air cavities located in facial and cranial bone lighten the skull and warm and moisten the air as it passes into the respiratory system.

Vertebral Column and Vertebrae (Slide 1 of 2)



Vertebral Column and Vertebrae

(Slide 2 of 2)

- **Vertebrae:** 26 bone segments from base of skull to tailbone in five divisions and separated by pads of cartilage called **intervertebral discs**
- **Cervical, thoracic, lumbar, sacral**
- **Coccyx:** tailbone

Bones: Thorax, Pelvis and Extremities

Thorax

- clavicle
- scapula
- sternum
- ribs

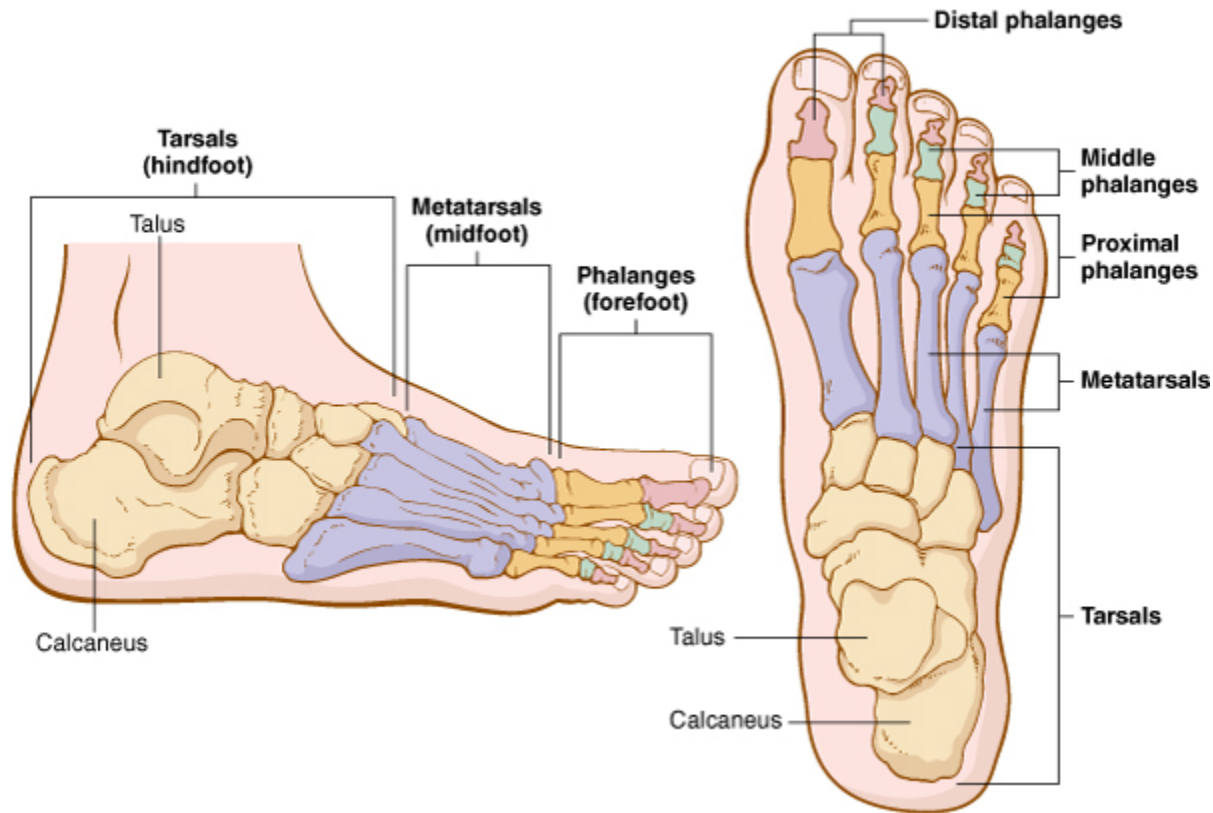
Pelvis

- pelvic girdle
- ilium
- ischium
- pubis

Arm and Hand

- humerus
- ulna
- radius
- carpals
- metacarpals
- phalanges

Bones of Leg and Foot



Bones of Leg and Foot and Associated Structures

- **Femur**
 - acetabulum
- **Patella**
- **Tibia**
 - medial malleolus
- **Fibula**
 - lateral malleolus
 - talus
- **Tarsals**
 - calcaneus
 - talus
- **Metatarsals**
- **Phalanges**

Vocabulary – Bones

(Slide 1 of 32)

Term
acetabulum
acromion

Vocabulary – Bones

(Slide 2 of 32)

Term	Meaning/Definition
acetabulum	Rounded depression in pelvis that joins the femur, forming the hip joint
acromion	Outward extension of shoulder blade forming the point of the shoulder

Vocabulary – Bones

(Slide 3 of 32)

Term
bone
bone depression
bone process

Vocabulary – Bones

(Slide 4 of 32)

Term	Meaning/Definition
bone	Dense, hard connective tissue composing the skeleton
bone depression	Opening or hollow region serving as a connection for bones, or passageway for blood vessels and nerves
bone process	Enlarged area that extends from bones as an attachment for muscles, tendons, and ligaments

Vocabulary – Bones

(Slide 5 of 32)

Term
calcium
cancellous bone
cartilage

Vocabulary – Bones

(Slide 6 of 32)

Term	Meaning/Definition
calcium	A mineral constituent of bone
cancellous bone	Spongy, porous bone tissue in inner part of bone
cartilage	Flexible, connective tissue

Vocabulary – Bones

(Slide 7 of 32)

Term
collagen
compact bone
cranial bones

Vocabulary – Bones

(Slide 8 of 32)

Term	Meaning/Definition
collagen	Dense, connective tissue protein strands found in bone and other tissues
compact bone	Hard, dense bone tissue, usually found around outer portion of bones
cranial bones	Skull bones: ethmoid, frontal, occipital, parietal, sphenoid, and temporal

Vocabulary – Bones

(Slide 9 of 32)

Term
diaphysis
disc
epiphyseal plate

Vocabulary – Bones

(Slide 10 of 32)

Term	Meaning/Definition
diaphysis	Shaft, or mid-portion, of a long bone
disc	Flat, round, plate-like structure
epiphyseal plate	Cartilaginous area at the ends of long bones where lengthwise growth takes in the immature skeleton

Vocabulary – Bones

(Slide 11 of 32)

Term
epiphysis
facial bones
fontanelle

Vocabulary – Bones

(Slide 12 of 32)

Term	Meaning/Definition
epiphysis	Each end of a long bone; area beyond the epiphyseal plate
facial bones	Bones of the face: lacrimal, mandibular, maxillary, nasal, vomer, and zygomatic
fontanelle	Soft spot (incomplete bone formation) between the skull bones of an infant

Vocabulary – Bones

(Slide 13 of 32)

Term
foramen magnum
haversian canals
ligament

Vocabulary – Bones

(Slide 14 of 32)

Term	Meaning/Definition
foramen magnum	Opening of the occipital bone through which the spinal cord passes
haversian canals	Minute spaces filled with blood vessels; found in compact bone
ligament	Fibrous connective tissue that binds bones to other

Vocabulary – Bones

(Slide 15 of 32)

Term
malleolus
manubrium
mastoid process

Vocabulary – Bones

(Slide 16 of 32)

Term	Meaning/Definition
malleolus	Round process on both sides of the ankle joint
manubrium	Upper portion of the sternum
mastoid process	Round projection on the temporal bone behind the ear

Vocabulary – Bones

(Slide 17 of 32)

Term
medullary cavity
metaphysis
olecranon

Vocabulary – Bones

(Slide 18 of 32)

Term	Meaning/Definition
medullary cavity	Central, hollowed-out area in the shaft of a long bone
metaphysis	Flared portion of a long bone, between the diaphysis and the epiphyseal plate
olecranon	Large process on the proximal end of the ulna

Vocabulary – Bones

(Slide 19 of 32)

Term
orthopedist
osseous tissue
ossification

Vocabulary – Bones

(Slide 20 of 32)

Term	Meaning/Definition
orthopedist	Medical doctor who specializes in bone, joint, and muscle conditions
osseous tissue	Bone tissue
ossification	Process of bone formation

Vocabulary – Bones

(Slide 21 of 32)

Term
ossification
osteoblast
osteoclast

Vocabulary – Bones

(Slide 22 of 32)

Term	Meaning/Definition
ossification	Process of bone formation
osteoblast	Bone cell that helps form bony tissue
osteoclast	Bone cell that absorbs and removes unwanted bony tissue

Vocabulary – Bones

(Slide 23 of 32)

Term
periosteum
phosphorus
physiatrist

Vocabulary – Bones

(Slide 24 of 32)

Term	Meaning/Definition
periosteum	Membrane surrounding bones; rich in blood vessels and nerve tissue
phosphorus	Mineral substance found in bones in combination with calcium
physiatrist	Medical doctor who specializes in rehabilitation

Vocabulary – Bones

(Slide 25 of 32)

Term
pubic symphysis
red bone marrow
ribs

Vocabulary – Bones

(Slide 26 of 32)

Term	Meaning/Definition
pubic symphysis	Area of confluence of the two pubic bones in the pelvis
red bone marrow	Found in cancellous bone; site of hematopoiesis
ribs	Twelve pairs of curved bones that form the chest wall; true ribs are the first 7 pairs; false ribs are pairs 8-10; floating ribs are pairs 11 and 12

Vocabulary – Bones

(Slide 27 of 32)

Term
sella turcica
sinus
styloid process

Vocabulary – Bones

(Slide 28 of 32)

Term	Meaning/Definition
sella turcica	Depression in the sphenoid bone where the pituitary gland is located
sinus	Hollow air cavity within a bone
styloid process	Pole-like process extending downward from the temporal bone on each side of the skull

Vocabulary – Bones

(Slide 29 of 32)

Term
suture
temporomandibular joint
tendon

Vocabulary – Bones

(Slide 30 of 32)

Term	Meaning/Definition
suture	Immovable joint between bones
temporomandibular joint	Connection on either side of the head between the temporal bone of the skull and the mandibular bone of the jaw
tendon	Fibrous connective tissue that binds muscles to bones

Vocabulary – Bones

(Slide 31 of 32)

Term
trabeculae
vertebra
xiphoid process
yellow bone marrow

Vocabulary – Bones

(Slide 32 of 32)

Term	Meaning/Definition
trabeculae	Supporting bundles of bony fibers in cancellous bone
vertebra	Individual segment of the spine composed of the vertebral body, vertebral arch, spinous process, transverse process, and lamina
xiphoid process	Lower, narrow portion of sternum
yellow bone marrow	Fatty tissue found in the medullary cavity of most adult long bones

QUICK QUIZ (Slide 1 of 4)

1. The knuckle-like process at the end of a bone near the joint...

A. fossa

B. condyle

C. tubercle

D. diaphysis

QUICK QUIZ (Slide 2 of 4)

2. The process of bone formation is

_____.

- A. olecranon
- B. ossification
- C. osteoblast
- D. xiphoid process

Terminology – Bones

Combining Forms (Slide 1 of 2)

Combining Form	Meaning
calc/o	calcium
calci/o	calcium
kyph/o	posterior curvature in thoracic region
lamin/o	lamina
lord/o	curve
lumb/o	loins, lower back

Terminology – Bones

Combining Forms (Slide 2 of 2)

Combining Form	Meaning
myel/o	bone marrow
orth/o	straight
oste/o	bone
scoli/o	crooked
spondyl/o	vertebra
vertebr/o	vertebra

Terminology – Bones

Suffixes

Suffix	Meaning
-blast	embryonic cell
-clast	to break
-listhesis	slipping
-malacia	softening
-physis	to grow
-porosis	pore, passage
-tome	instrument to cut

Terminology Related to Specific Bones

Combining Forms (Slide 1 of 5)

Combining Form	Meaning
acetabul/o	acetabulum (hip socket)
calcane/o	calcaneus (heel)
carp/o	carpals (wrist bones)
clavicul/o	clavicle (collar bone)
cost/o	ribs
crani/o	cranium (skull)

Terminology Related to Specific Bones

Combining Forms (Slide 2 of 5)

Combining Form	Meaning
femor/o	femur (thigh bones)
fibul/o	fibula (smaller lower leg bone)
humer/o	humerus (upper arm bone)
ili/o	ilium (upper part of pelvic bone)
ischi/o	ischium (part of pelvic bone)
malleol/o	malleolus

Terminology Related to Specific Bones

Combining Forms (Slide 3 of 5)

Combining Form	Meaning
mandibul/o	mandible (lower jaw bone)
maxill/o	maxilla (upper jaw bone)
metacarp/o	metacarpals (hand bones)
metatars/o	metatarsals (foot bones)
olecran/o	olecranon (elbow)
patell/o	patella (kneecap)
pelv/i	pelvis

Terminology Related to Specific Bones

Combining Forms (Slide 4 of 5)

Combining Form	Meaning
perone/o	fibula
phalang/o	phalanges (finger, toe bones)
pub/o	pubis
radi/o	radius
scapul/o	scapula (shoulder blade)

Terminology Related to Specific Bones

Combining Forms (Slide 5 of 5)

Combining Form	Meaning
stern/o	sternum (breastbone)
tars/o	tarsals
tibi/o	tibia (shin bone)
uln/o	ulna (lower arm bone)

Pathology — Bones

- **Ewing sarcoma:** malignant bone tumor
- **exostosis:** bony growth on surface of bone
- **osteogenic sarcoma:** malignant tumor arising from bone
- **osteomalacia:** softening of bone
- **osteomyelitis:** inflammation secondary to infection
- **talipes:** congenital abnormality of hindfoot involving the talus (clubfoot)

Pathologic Conditions

Fractures of Bones

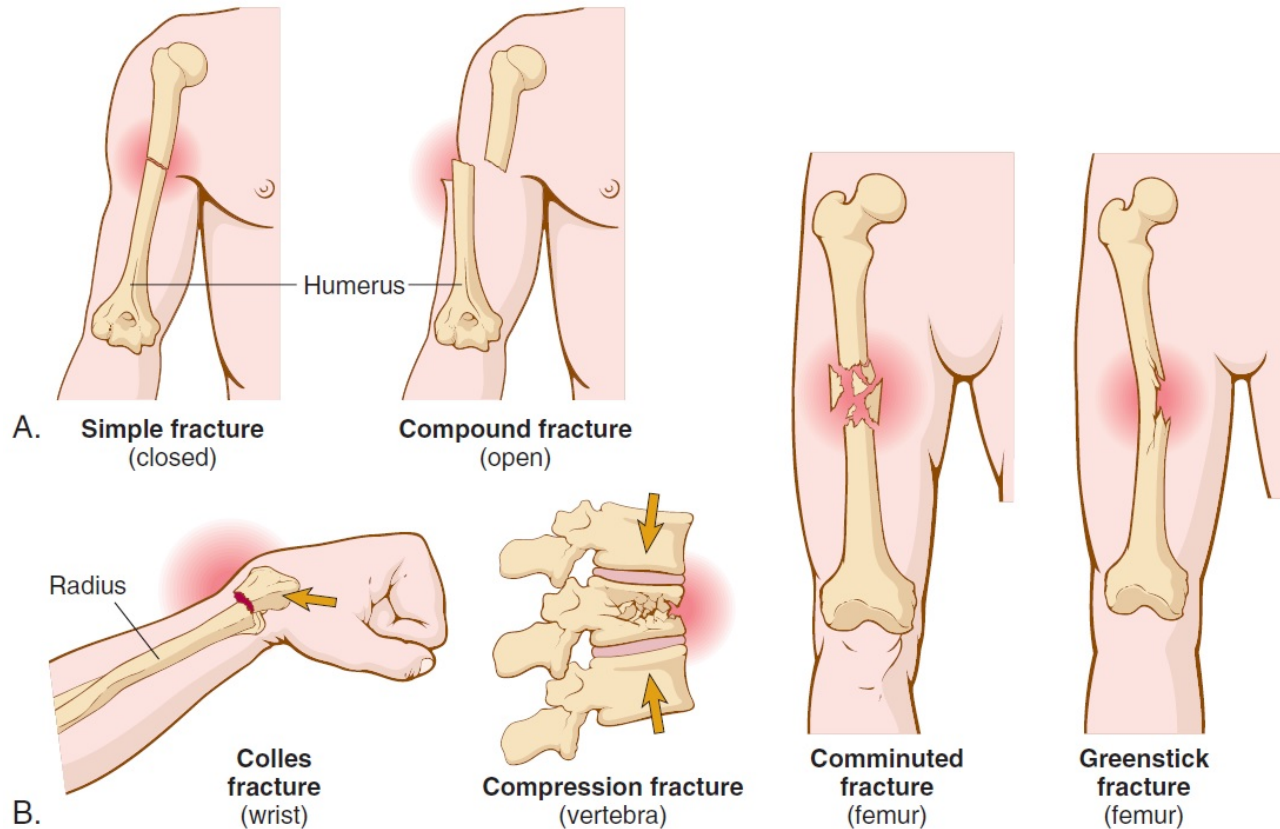


FIGURE 15-14 A, Simple and compound fractures. B, Types of fractures. A pathologic fracture (not pictured) is caused by disease of bones (tumor, infection, osteoporosis) that causes them to weaken. Cartilage is found at the site of injury when a bone breaks and is in the process of healing.

Chapter 15

Lesson 15.2

Joints (Slide 1 of 2)

- **Joint (articulation):** coming together of two or more bones
 - **Suture joints:** immovable (skull)
 - **Synovial joints:** freely movable (ball and socket types, e.g., hip or shoulder), (hinge type, for example, elbow, knee, ankle)

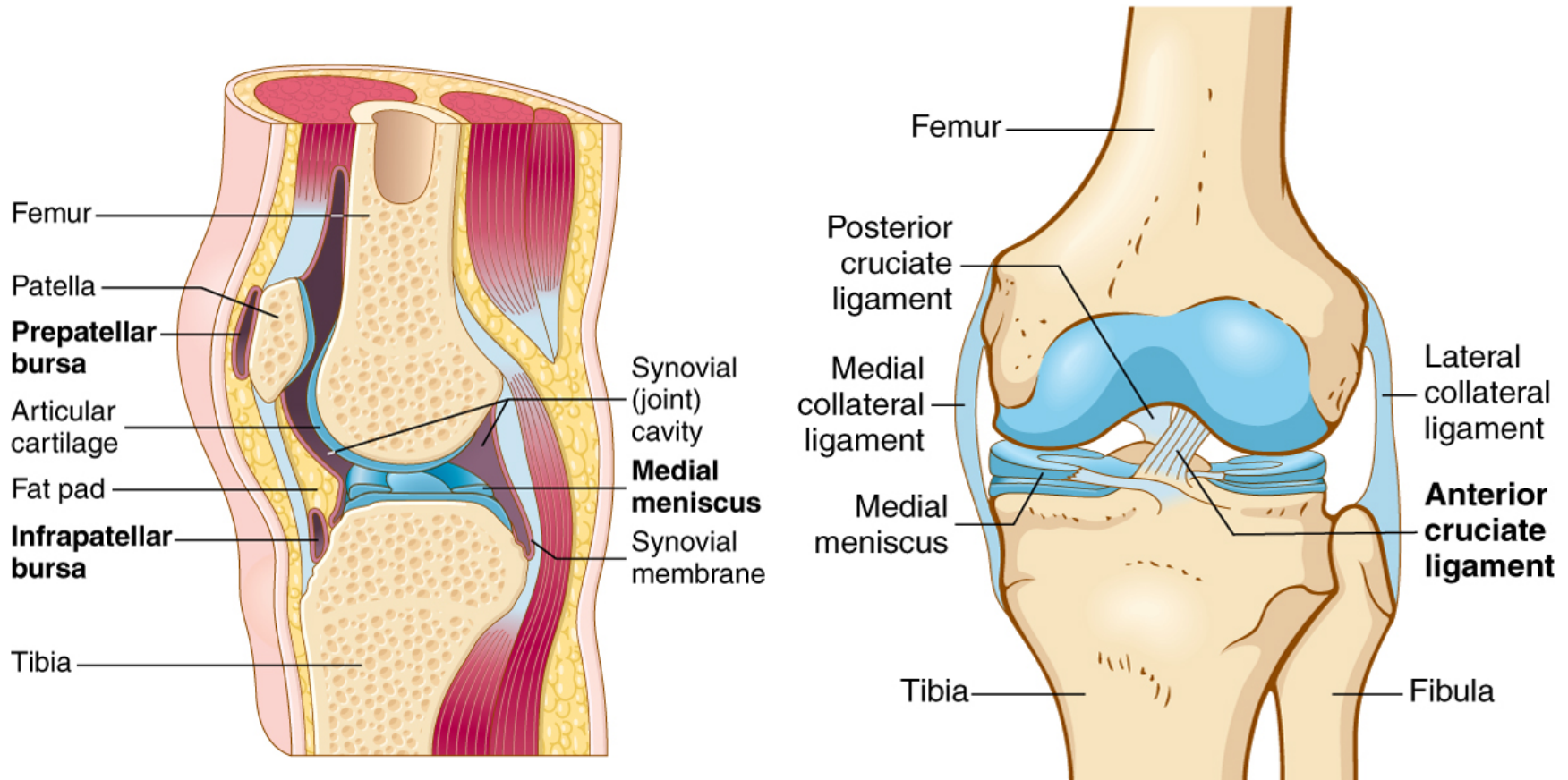
Synovial Joints

- Joint capsule surrounds bone.
- Ligaments band bones together.
- Bones are covered by articular cartilage.
- Synovial membrane lies under capsule and lines synovial cavity; filled with synovial fluid.
- A meniscus is a crescent-shaped fibrocartilaginous structure that partly divides a joint cavity; acts as a protective cushion; present in the knee.

Bursae

- Closed sacks of synovial fluid with a synovial membrane located near, but not within a joint. Needed where sliding must take place.
- Common sites:
 - between tendons (connective tissue connecting muscle to bone) and bones
 - between ligaments (binding bone to bone) and bones
 - between skin and bones (with prominent bony anatomy)

Joints (Slide 2 of 2)



Vocabulary – Joints (Slide 1 of 6)

Term
articular cartilage
articulation
bursa (<i>plural:</i> bursae)
ligament

Vocabulary – Joints (Slide 2 of 6)

Term	Meaning/Definition
articular cartilage	Smooth, glistening white tissue that covers the surface of a joint
articulation	Any type of joint
bursa (<i>plural: bursae</i>)	Sac of fluid near a joint
ligament	Fibrous connective bands binding bones to other bones

Vocabulary – Joints (Slide 3 of 6)

Term
meniscus
suture joint
synovial cavity

Vocabulary – Joints (Slide 4 of 6)

Term	Meaning/Definition
meniscus	Crescent-shaped fibrocartilaginous structure found in the knee
suture joint	Immovable joint
synovial cavity	Space between bones at a synovial joint

Vocabulary – Joints (Slide 5 of 6)

Term
synovial fluid
synovial joint
synovial membrane
tendon

Vocabulary – Joints (Slide 6 of 6)

Term	Meaning/Definition
synovial fluid	Viscous fluid within the synovial cavity
synovial joint	Freely movable joint
synovial membrane	Tissue lining the synovial cavity; produces synovial fluid
tendon	Fibrous connective bands that connect muscles to bones

QUICK QUIZ (Slide 3 of 4)

3. The fibrous connective band that binds muscles to bones is

_____.

- A. ligament
- B. articulation
- C. synovial membrane
- D. tendon

Combining Forms and Terminology

Joints (Slide 1 of 2)

Combining Form	Meaning
ankyl/o	stiff
arthr/o	joint
articul/	joint
burs/o	bursa
chondr/o	cartilage

Combining Forms and Terminology

Joints (Slide 2 of 2)

Combining Form	Meaning
ligament/o	ligament
rheumat/o	watery flow
synov/o	synovial membrane
ten/o	tendon
tendin/o	tendon

Suffixes and Terminology

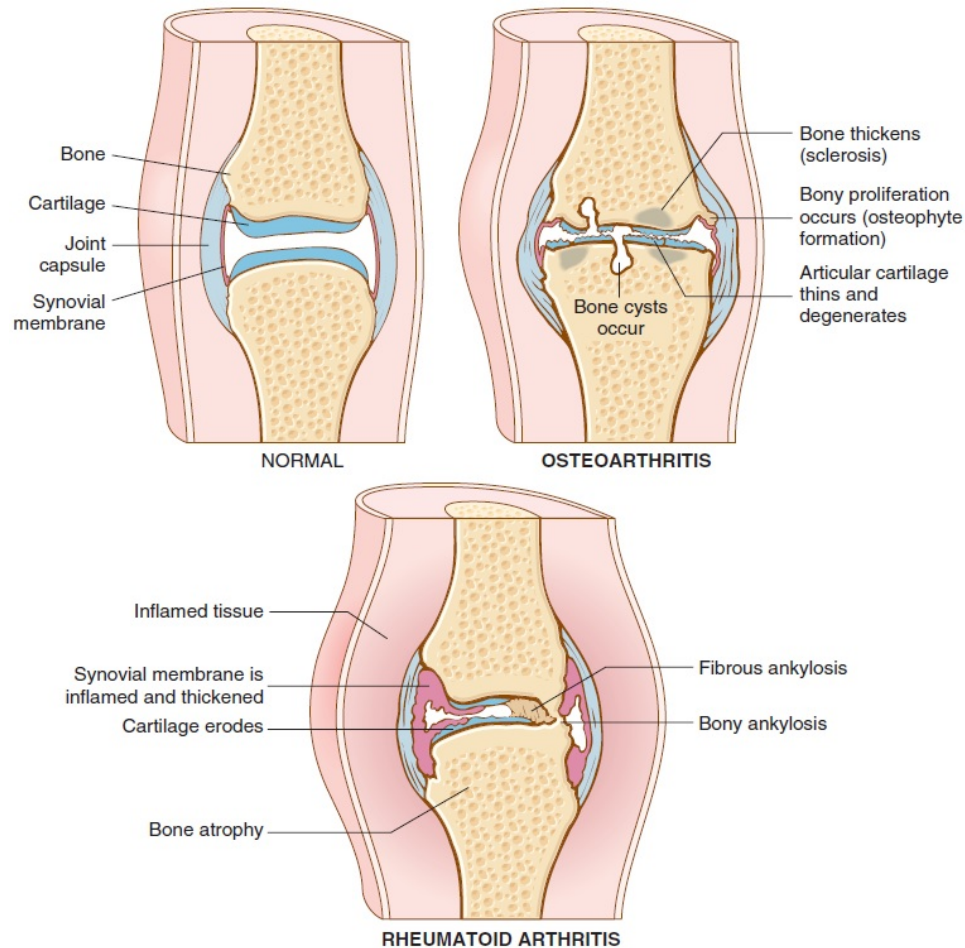
Joints

Combining Form	Meaning
-desis	to bind, tie together
-stenosis	narrowing

Pathology — Joints

- **Arthritis:** inflammation of any joint
 - **Ankylosing spondylitis:** chronic progressive stiffening of joints, mostly spine
 - **Gouty arthritis:** inflammation due to excessive uric acid in body
 - **Osteoarthritis (OA):** loss of articular cartilage and formation of bone spurs at articular surfaces; can occur in any joint, but mainly in spine, hips, and knees of older people
 - **Rheumatoid arthritis (RA):** autoimmune reaction against joint tissues (synovial membrane)

Osteoarthritis and Rheumatoid Arthritis

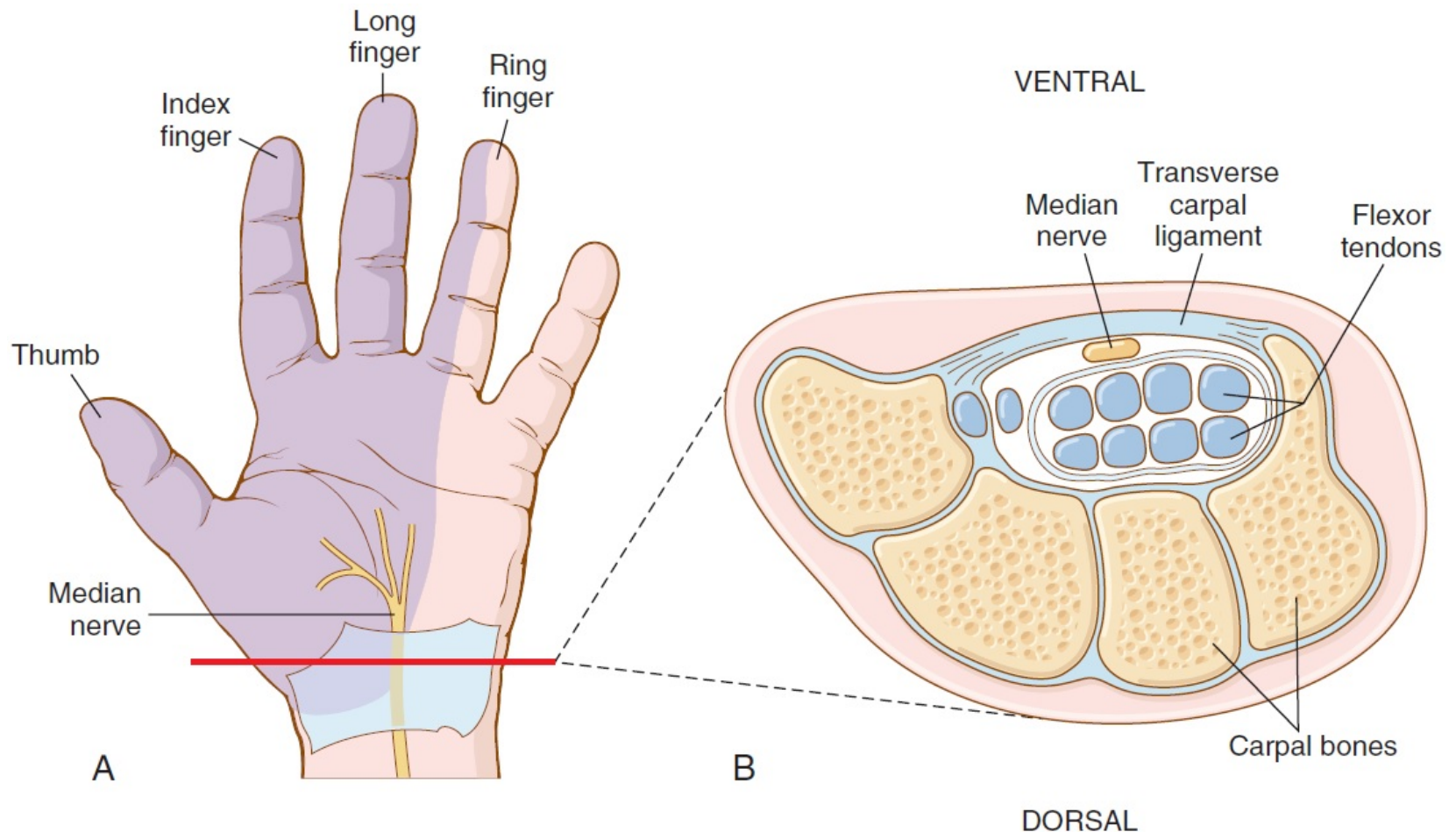


Pathology — Joints (Slide 1-4)

- **Bunion:** swelling of medial aspect of joint between big toe and first metatarsal bone
- **Carpal tunnel syndrome:** compression of median nerve as it passes between transverse ligament and bones and tendons of the wrist
- **Dislocation:** displacement of bone from a joint

Pathology — Joints (Slide 2 of 4)

Carpal Tunnel Syndrome



Pathology — Joints (Slide 3 of 4)

- **Ganglion:** a fluid-filled cyst arising from the joint capsule or a tendon in the wrist.
- **Herniation of an intervertebral disc:** abnormal protrusion of the disc into the neural canal or against spinal nerves; “slipped disc”.
- **Lyme disease:** recurrent arthritis, myalgia, malaise and neurologic and cardiac symptoms; often marked by a “bull’s eye” rash at the site of the tick bite.

Pathology — Joints (Slide 4 of 4)

- **Sprain:** trauma to joint with pain, swelling and injury to ligaments without rupture
- **Systemic lupus erythematosus (SLE):** chronic inflammatory disease involving joints, skin, kidneys, nervous system, heart and lungs affecting collagen in tendons, ligaments, bones, and cartilage all over the body

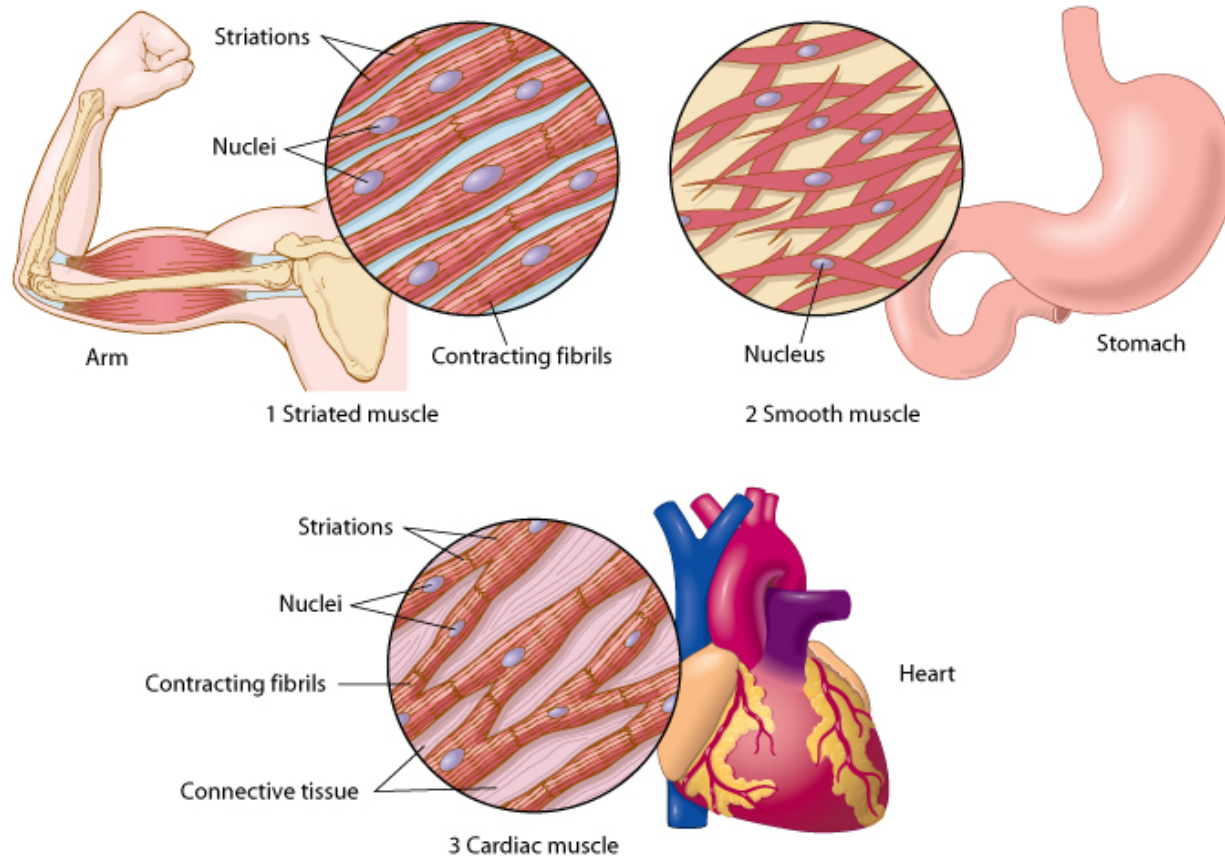
Chapter 15

Lesson 15.3

Muscles

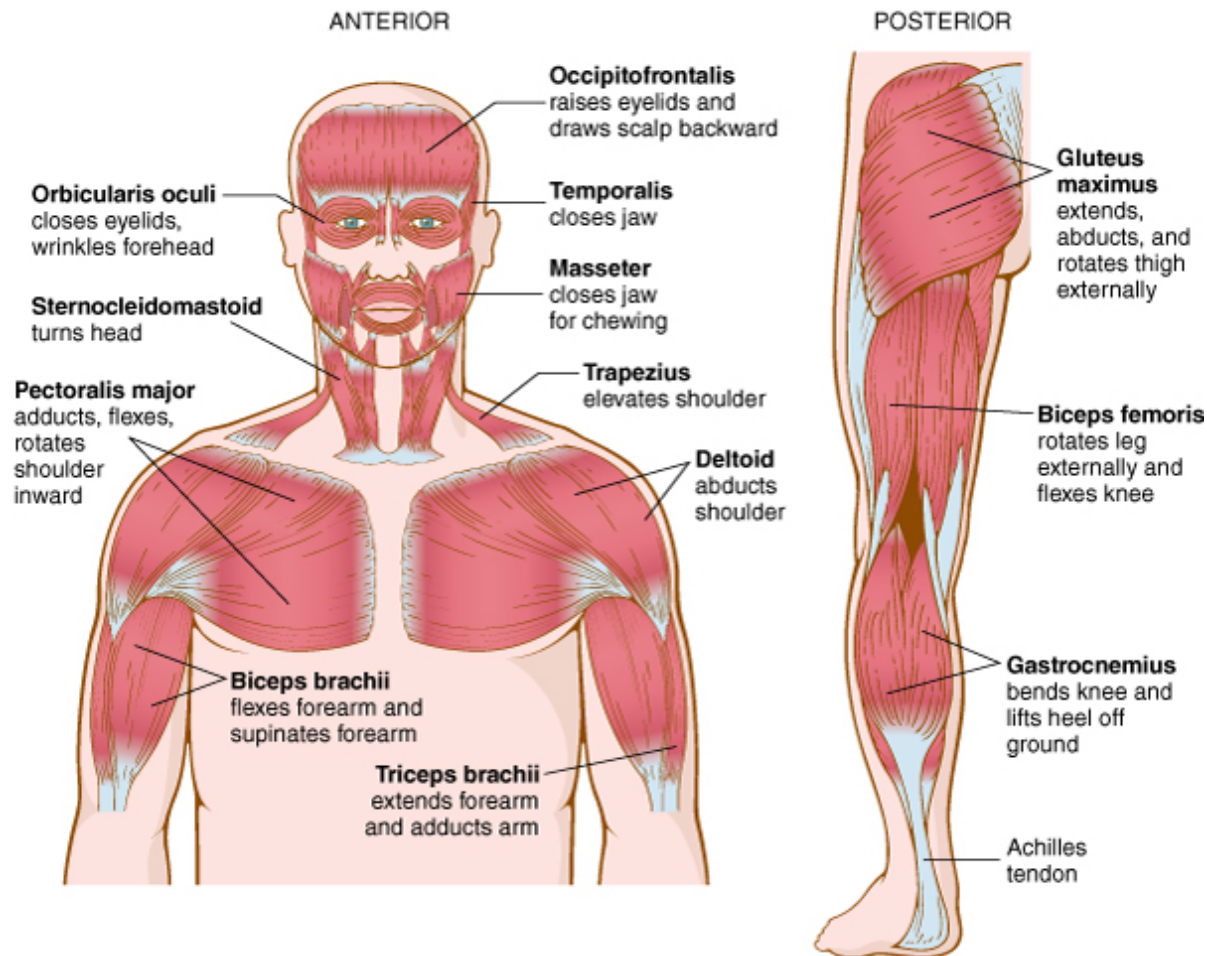
- Three types of muscles: striated, smooth, cardiac
- **Striated:** under voluntary control—they move all the bones as well as face and eyes
- **Smooth:** control is involuntary—they move internal organs (digestive tract, blood vessels, ducts of glands)
- **Cardiac:** not consciously controlled and found exclusively in the heart

Types of Muscle



Actions of Skeletal Muscles

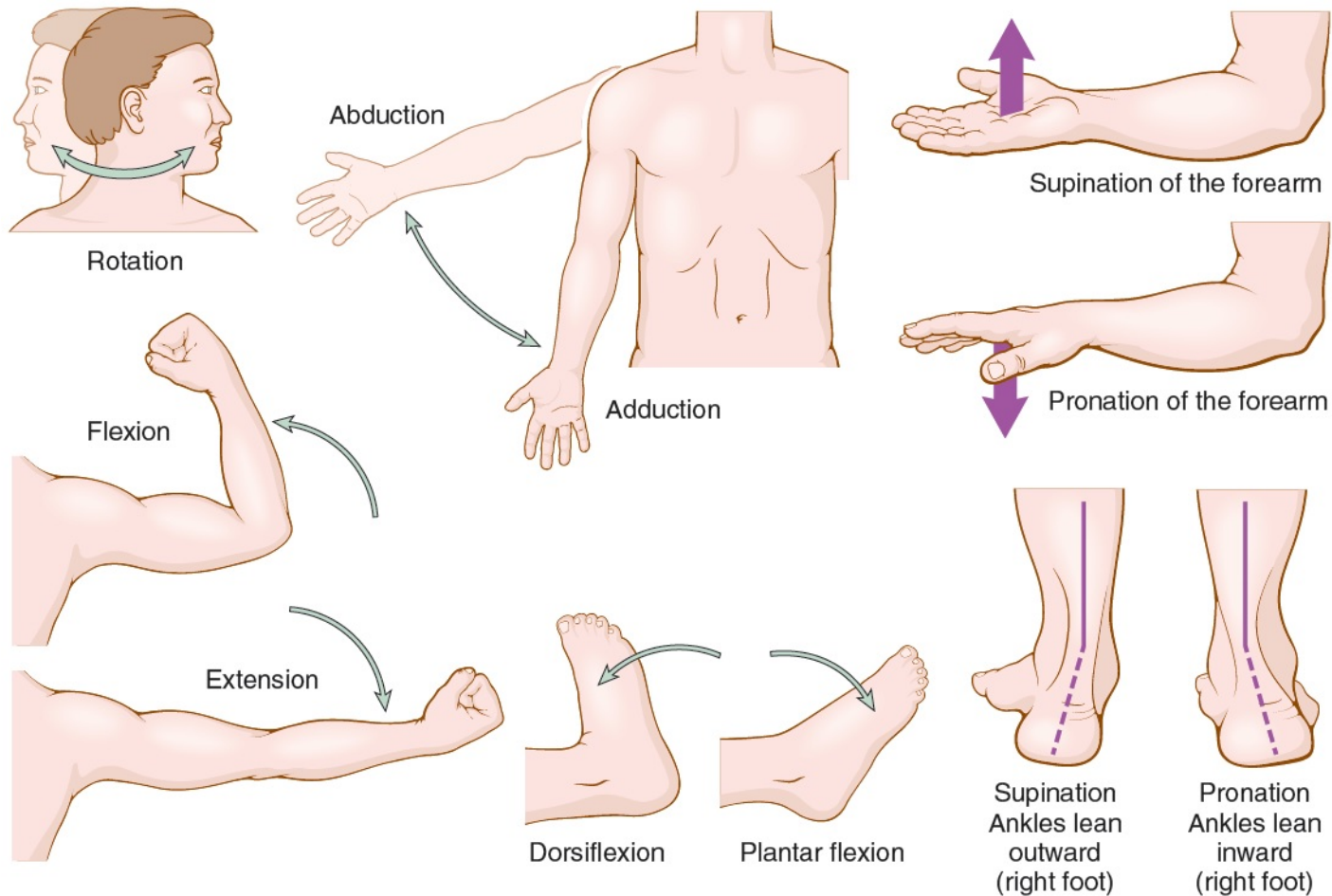
(slide 1 of 3)



Actions of Skeletal Muscles (slide 2 of 3)

- flexion
- extension
- abduction
- adduction
- rotation
- dorsiflexion
- plantar flexion
- supination
- pronation

Actions of Muscles (slide 3 of 3)



Vocabulary – Muscles (Slide 1 of 8)

Term
abduction
adduction
dorsiflexion
extension

Vocabulary – Muscles (Slide 2 of 8)

Term	Meaning/Definition
abduction	Movement away from midline of body
adduction	Movement toward midline of body
dorsiflexion	Upper movement of the foot
extension	Straightening of a flexed limb

Vocabulary – Muscles (Slide 3 of 8)

Term
fascia
flexion
insertion of a muscle
origin of a muscle

Vocabulary – Muscles (Slide 4 of 8)

Term	Meaning/Definition
fascia	Fibrous membrane separating and enveloping muscles
flexion	Downward movement of the foot
insertion of a muscle	Connection of the muscle to a bone that moves
origin of a muscle	Connection of the muscle to a stationary bone

Vocabulary – Muscles (Slide 5 of 8)

Term
plantar flexion
pronation
rotation
skeletal muscle

Vocabulary – Muscles (Slide 6 of 8)

Term	Meaning/Definition
plantar flexion	Bending the sole of the foot downward toward the ground
pronation	Turning the palm downward
rotation	Circular movement around a central point
skeletal muscle	Muscle connected to bones; voluntary or striated muscle

Vocabulary – Muscles (Slide 7 of 8)

Term
smooth muscle
striated muscle
supination
visceral muscle

Vocabulary – Muscles (Slide 8 of 8)

Term	Meaning/Definition
smooth muscle	Visceral muscle
striated muscle	Skeletal muscle
supination	Turning the palm upward
visceral muscle	Muscle connected to internal organs; involuntary or smooth muscle

QUICK QUIZ (Slide 4 of 4)

4. Movement *AWAY* from the midline of the body is _____.

- A. adduction
- B. abduction
- C. flexion
- D. dorsiflexion

Combining Forms and Terminology

Muscles (Slide 1 of 2)

Combining Form	Meaning
fasci/o	fascia
fibr/o	fibrous
leiomy/o	smooth muscle
my/o	muscle

Combining Forms and Terminology

Muscles (Slide 2 of 2)

Combining Form	Meaning
myocardi/o	heart muscle
myos/o	muscle
plant/o	sole of the foot
rhabdomy/o	skeletal muscle connected to bones
sarco	muscles and flesh

Suffixes and Terminology

Muscles

Suffix	Meaning
-asthenia	lack of strength
-trophy	development, nourishment

Prefixes and Terminology

Muscles

Prefix	Meaning
ab-	away from
ad-	toward
dorsi-	back
poly-	many, much

Pathology — Muscles

- **Muscular dystrophy:** group of inherited diseases with progressive muscle weakness and degeneration without nervous system involvement
- **Polymyositis:** chronic inflammatory myopathy; may be an autoimmune disorder

Laboratory Tests

- Antinuclear antibody test (ANA)
- Erythrocyte sedimentation rate (ESR)
- Rheumatoid factor test (RF)
- Serum calcium (Ca)
- Serum creatine kinase (CK)
- Uric acid test

Clinical Procedures Related to Bones, Joints, and Muscle

- Arthrocentesis
- Arthrography
- Arthroplasty
- Arthroscopy
- Bone density test
- Bone scan
- Computed tomography (CT)
- Discography
- Electromyography (EMG)
- Magnetic resonance imaging (MRI)
- Muscle biopsy

Abbreviations (Slide 1 of 6)

Prefix

AC

ACL

ANA

BKA

BMD

C1 TO C7

Ca

CK

CMC

CTS

DEXA or DXA

DMARD

Abbreviations (Slide 2 of 6)

Prefix	Meaning
AC	Acromioclavicular (joint)
ACL	Anterior cruciate ligament of the knee
ANA	Antinuclear antibody
BKA	Below-knee amputation
BMD	Bone mineral density
C1 TO C7	Cervical vertebrae
Ca	Calcium
CK	Creatine kinase
CMC	Carpometacarpal (joint)
CTS	Carpal tunnel syndrome
DEXA or DXA	Dual-energy x-ray absorptiometry
DMARD	Disease-modifying antirheumatic drug

Abbreviations (Slide 3 of 6)

Prefix

DO

DTRs

EMG

ESR (sed rate)

HNP

IM

L1 to L5

NSAID

OA

ORIF

ortho

Abbreviations (Slide 4 of 6)

Prefix	Meaning
DO	Doctor of osteopathy
DTRs	Deep tendon reflexes
EMG	Electromyography
ESR (sed rate)	Erythrocyte sedimentation rate
HNP	Herniated nucleus pulposus
IM	Intramuscular
L1 to L5	Lumbar vertebrae
NSAID	Nonsteroidal anti-inflammatory drug
OA	Osteoarthritis
ORIF	Open reduction (of fracture)/ internal fixation
ortho	Orthopedics (or orthopaedics)

Abbreviations (Slide 5 of 6)

Prefix

OT

P

PT

RA

RF

ROM

SLE

T1 to T12

TKR

THR

TMJ

Abbreviations (Slide 6 of 6)

Prefix	Meaning
OT	Occupational therapy
P	Phosphorus
PT	Physical therapy
RA	Rheumatoid arthritis
RF	Rheumatoid factor
ROM	Range of motion
SLE	Systemic lupus erythematosus
T1 to T12	Thoracic vertebrae
TKR	Total knee replacement/arthroplasty
THR	Total hip replacement/arthroplasty
TMJ	Temporomandibular joint